8.4 UAV Applications in Smart Agriculture

The integration of unmanned aerial vehicles (UAVs) into agricultural systems has significantly transformed the way farming operations are planned, monitored, and executed. Driven by the need to increase productivity while reducing resource consumption, UAV-based precision agriculture offers highly accurate, data-driven insights that enable farmers to make informed decisions at field, crop, and plant scales. These systems leverage aerial imaging, environmental sensing, and intelligent data analytics to enhance crop performance, ensure efficient use of inputs, and support sustainable agricultural practices. As global food demand continues to rise, UAVs play a crucial role in bridging the gap between traditional cultivation methods and digitally optimized farming systems.

Table 8.4 highlights the key technological components of UAV systems used in smart agriculture and explains how each contributes to improving agricultural productivity, decision-making, and resource efficiency. The table shows that UAVs are not merely aerial imaging platforms, but *integrated data-driven farm management tools*. For instance, *RGB cameras* allow farmers to visually inspect crop conditions and detect abnormalities quickly, while *multispectral sensors* provide deeper insights into crop health by analyzing plant vigor, stress levels, and nutrient status information that is not visible to the naked eye. *Thermal imaging sensors* further expand monitoring capabilities by identifying water stress and guiding irrigation scheduling, thus supporting water conservation efforts.

Table 8.4 UAV technologies and their functional benefits in smart agriculture.⏎

UAV component/feature	Function in agriculture	Benefit to farmers
RGB cameras	Visual crop health inspection and	Quick field assessment; early detection of